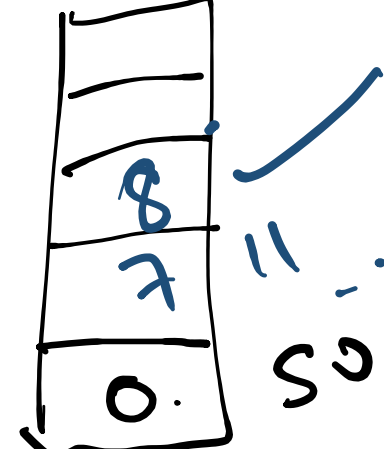
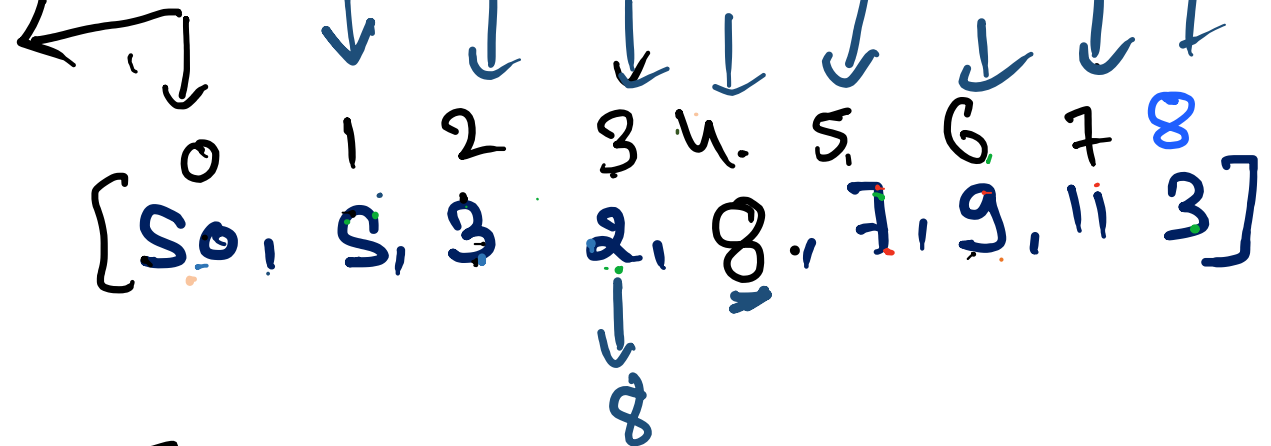
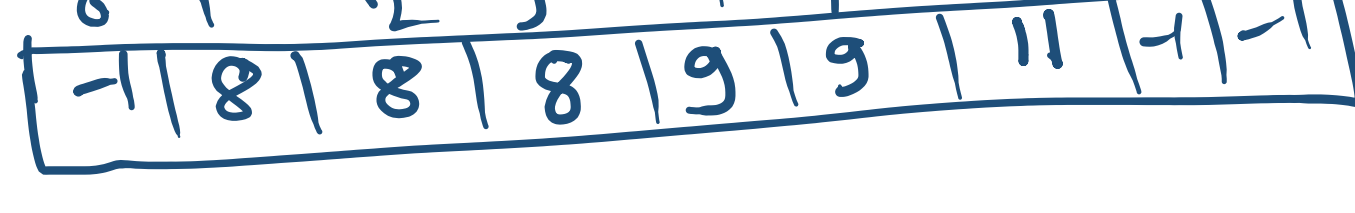
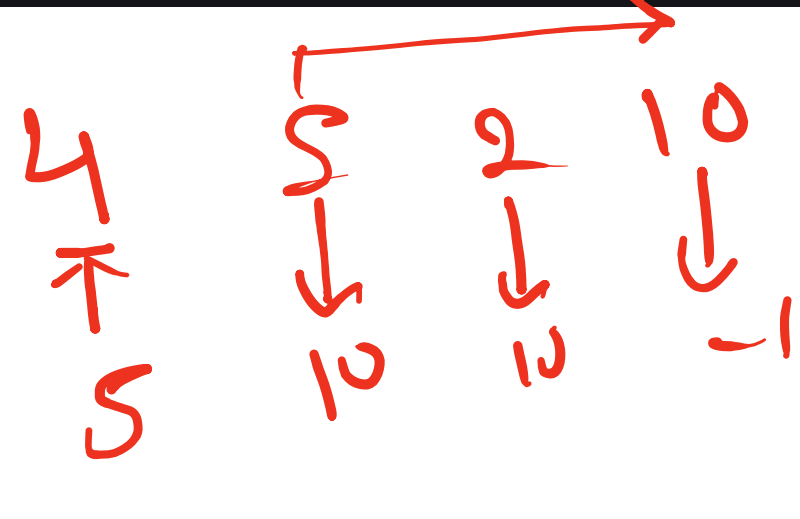


Next Greater Element

Given an array of integers, for each element in the array, find the **next greater element (NGE)** in the array. The next greater element for an element x is the first element y to the right of x in the array such that $y > x$. If no such element exists, the NGE for that element is -1.

45210



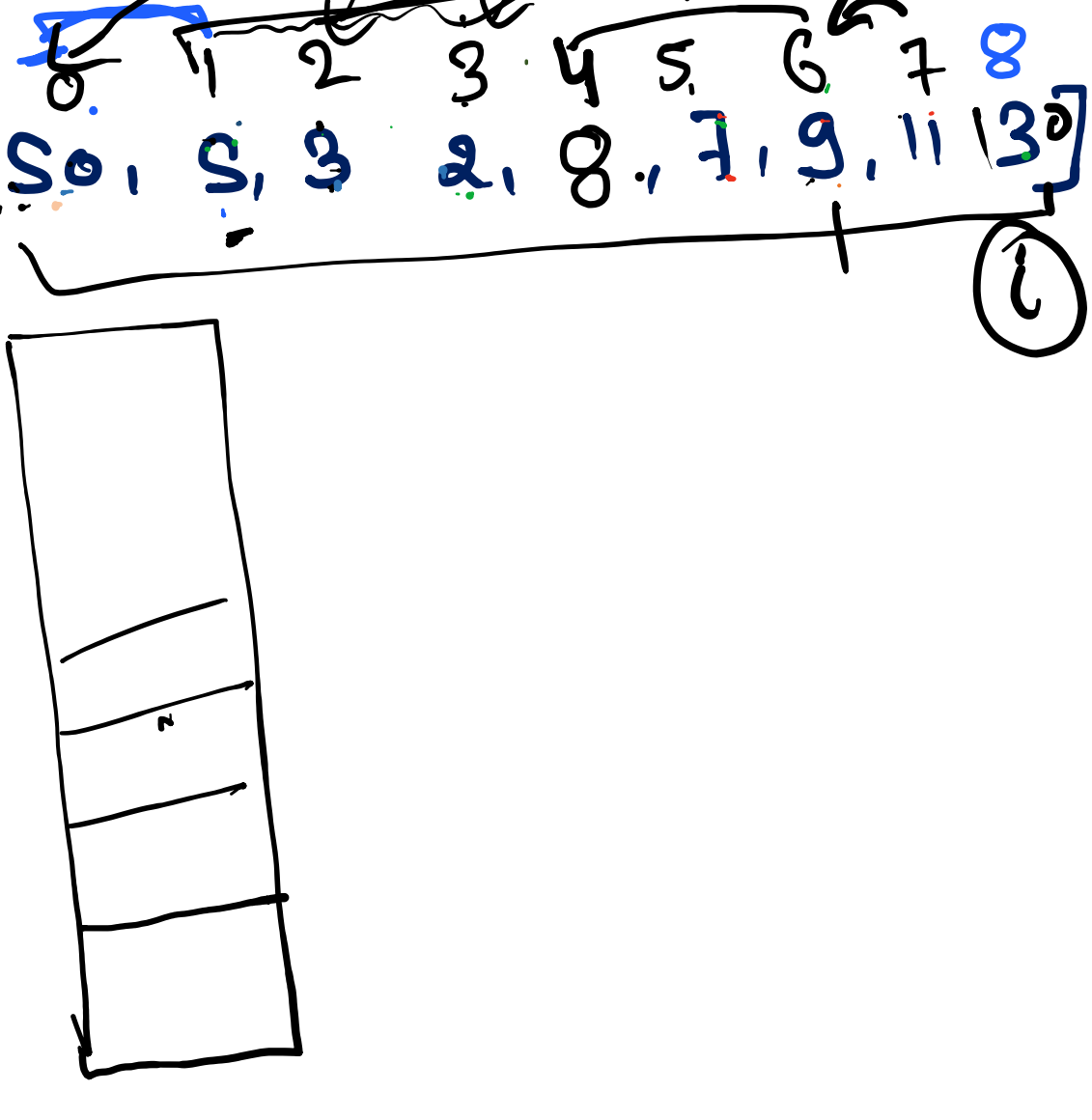
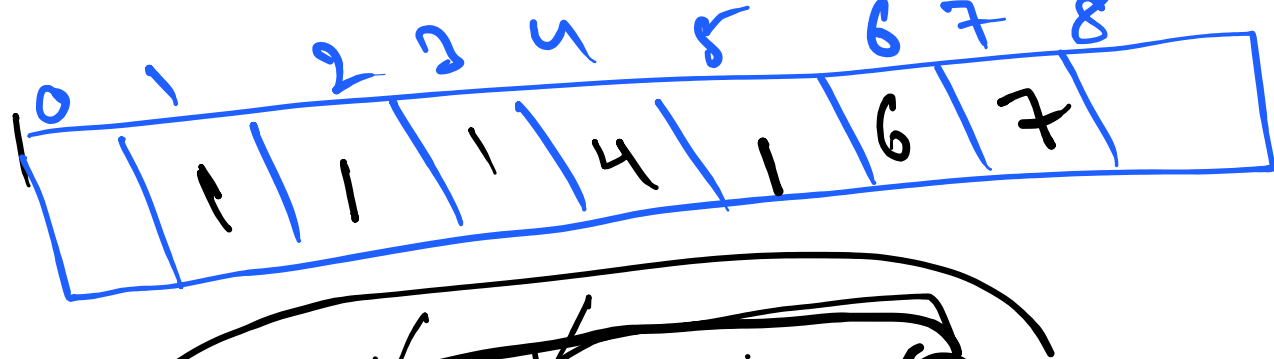
```
for (i = 0; i < arr.length; i++) {
    while (!st.isEmpty() && arr[i] > arr[st.peek()]) {
        ans[st.pop()] = arr[i];
    }
    st.push(i);
}
```

```
while (!st.isEmpty()) {
    ans[st.pop()] = -1;
}
```

30 35 40 38 35

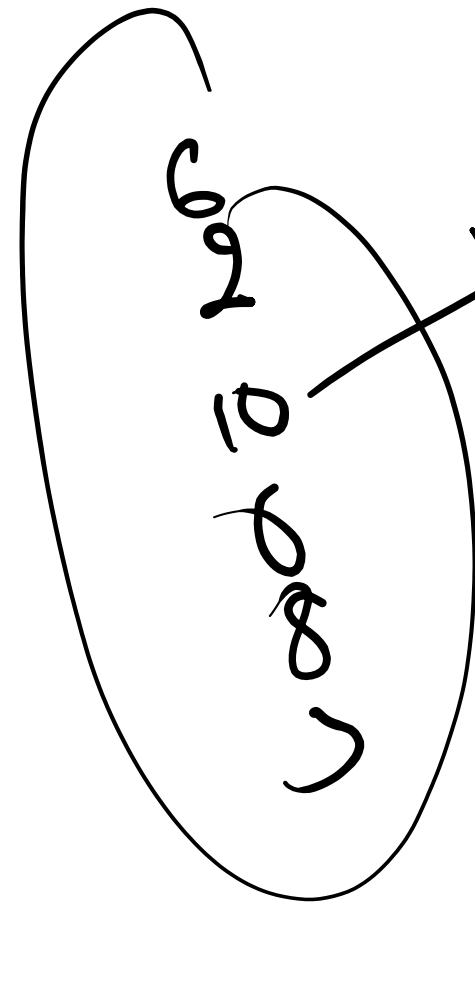
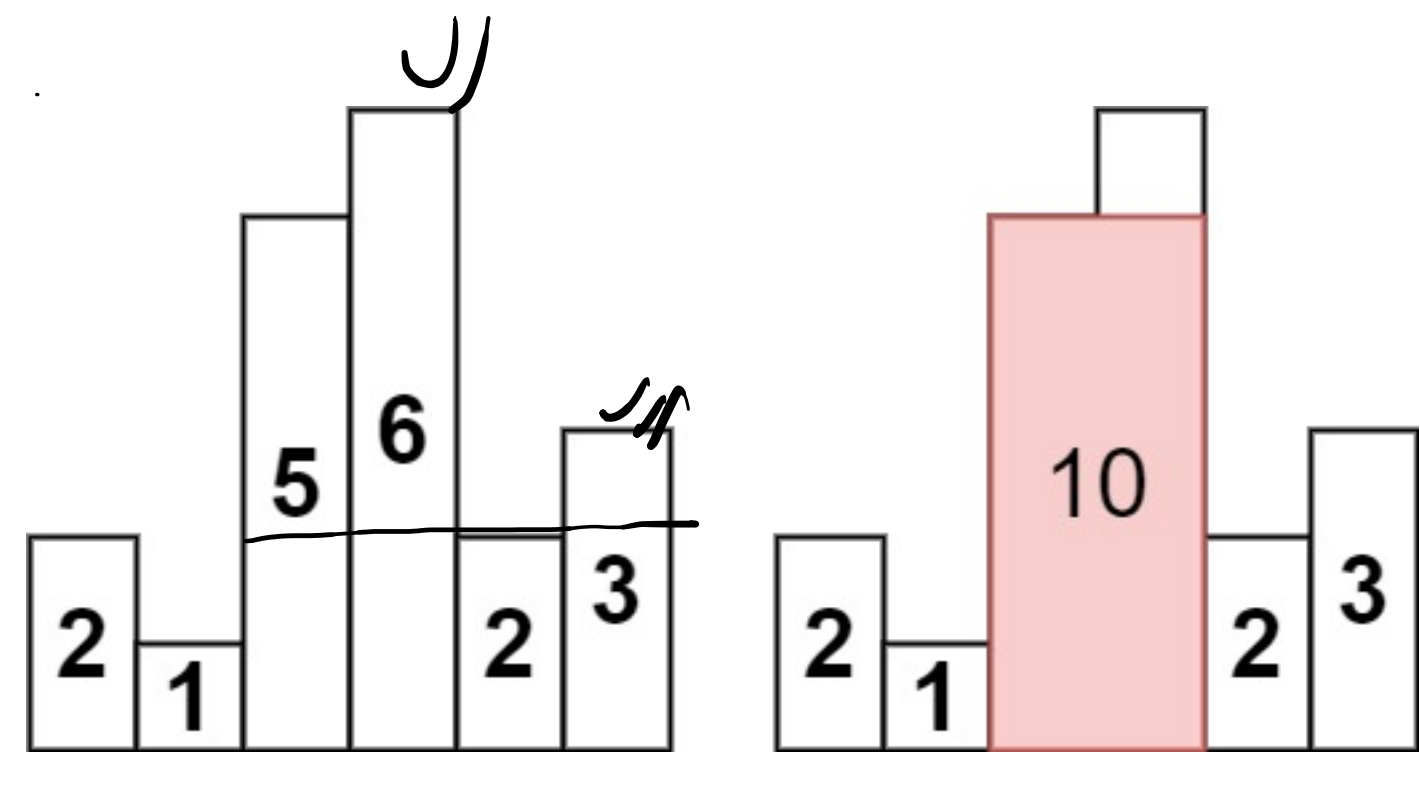


array

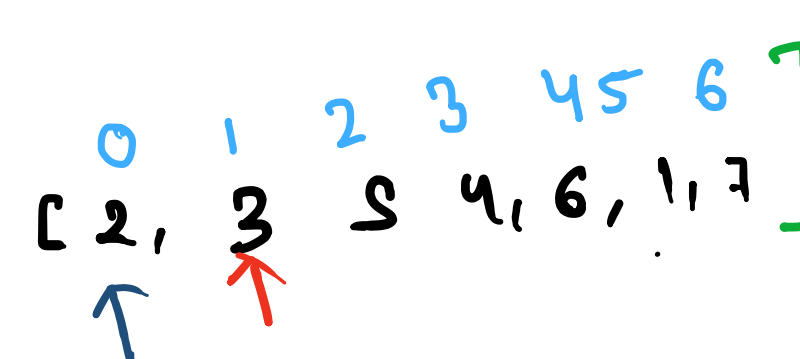


array

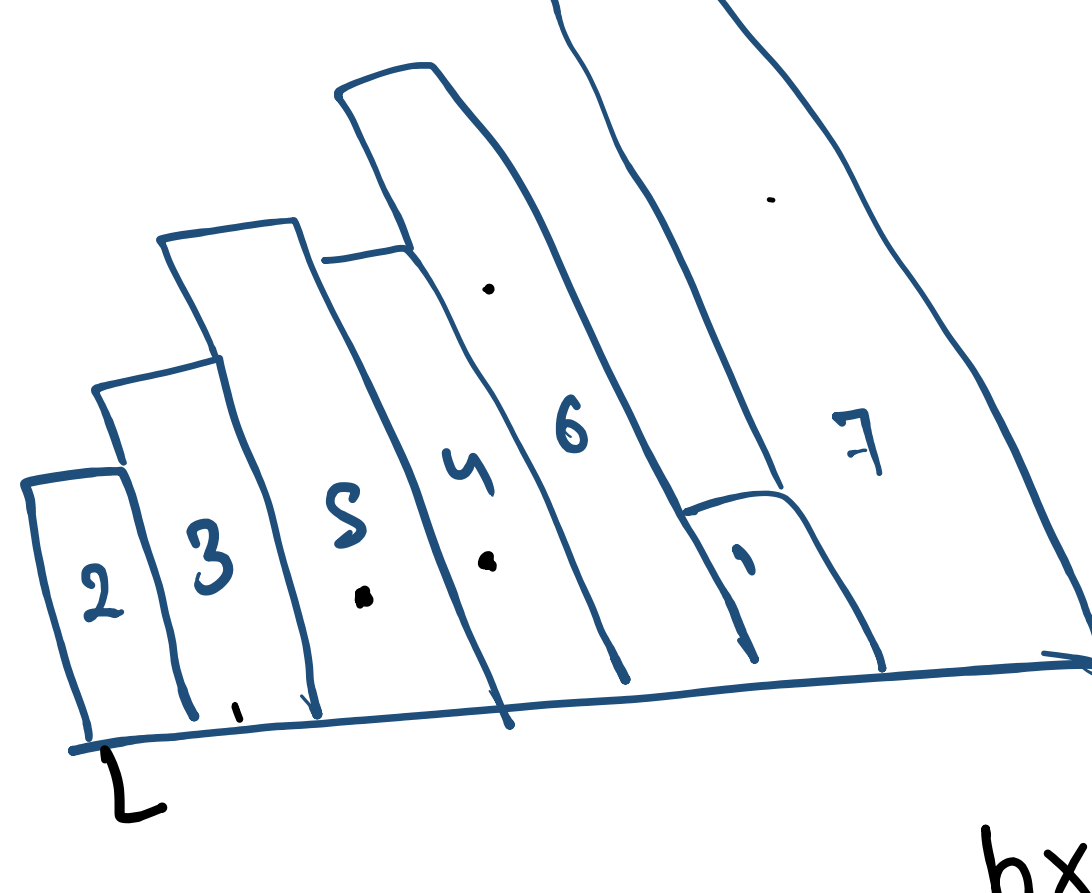
```
for (i = 0; i < arr.length; i++) {
    while (!st.isEmpty() && arr[i] > arr[st.peek()]) {
        ans[st.pop()] = arr[i];
    }
    st.push(i);
}
```



10-2=8

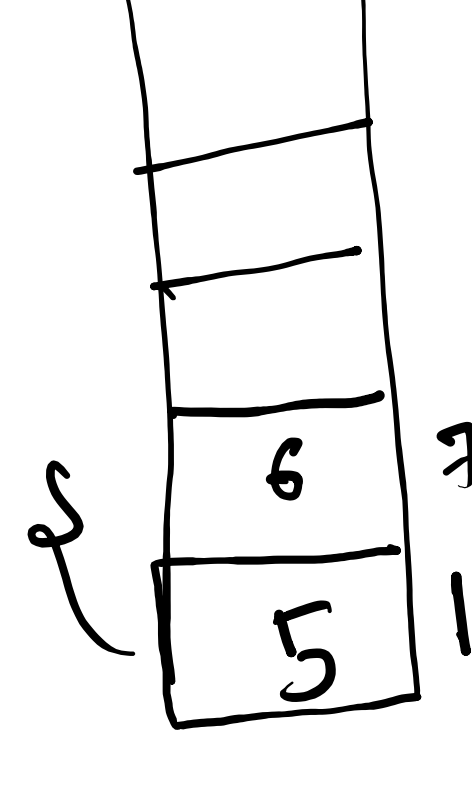
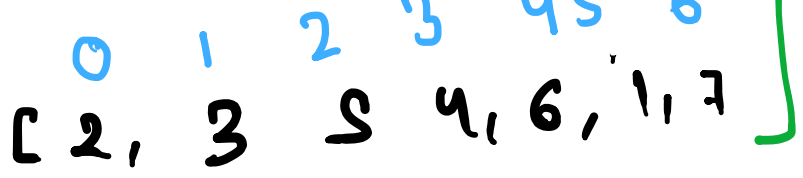


2x5=10
3x4=12
5x1=5
4x3=12
6x1=6
1x7=7
2x8=16



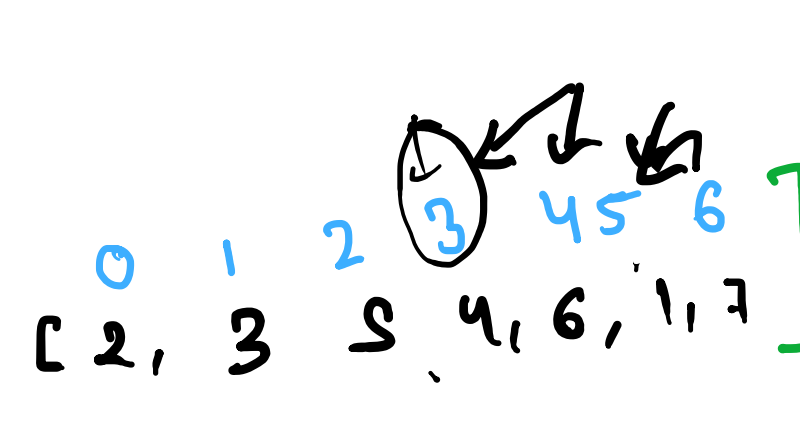
12

h x (R-L-1)



arr[i] > arr[st.peek()]
h x f

5 x (0-1-1) = 5
6 x (5-3-1) = 6
4 x (5-1-1) = 12
3 x (5-0-1) = 12
2 x 8 = 16



ans = 5, 8, 12

h = 5, R = 3, L = 1
5 x (3-1-1) = 5
R = 5
6 x (5-3-1) = 6
4 x (5-1-1) = 12
2 x 8 = 16

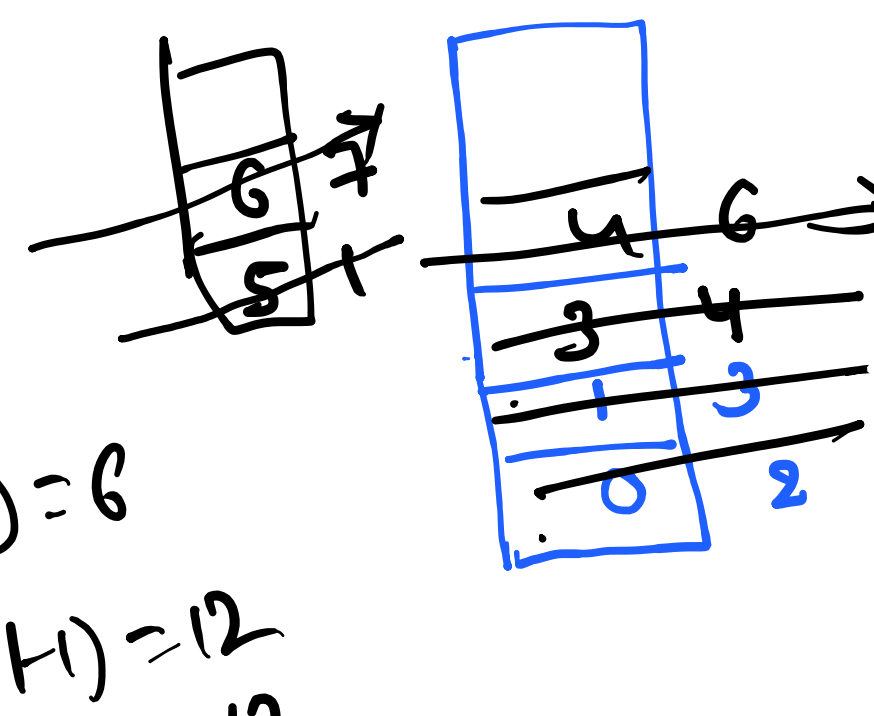
```
for (i = 0; i < arr.length; i++) {
    while (!st.isEmpty() && arr[i] < arr[st.peek()]) {
        R = i;
        h = arr[st.pop()];
        if (st.isEmpty()) {
            ans = Math.max(ans, h * R);
        } else {
            L = st.peek();
            ans = Math.max(ans, h * (R - L - 1));
        }
    }
    st.push(i);
}
```

```
public static int LargestRectangle(int[] arr) {
    Stack<Integer> st = new Stack<>();
    int ans = 0;
    for (int i = 0; i < arr.length; i++) {
        while (!st.isEmpty() && arr[i] < arr[st.peek()]) {
            int r = i;
            int h = arr[st.pop()];
            if (st.isEmpty()) {
                ans = Math.max(ans, h * r);
            } else {
                int l = st.peek();
                ans = Math.max(ans, h * (r - l - 1));
            }
        }
        st.push(i);
    }
}
```

ans = 8, 8, 12



7 x (3-5-1) = 6
1 x (7) = 7



5, 6 x (5-3-1) = 6
6, 4 x (5-1-1) = 12
12, 3 x (5-0-1) = 12
12, 2 x 8 = 16

